

CO1-AT2-CONCEPT MAPPING QUESTIONS

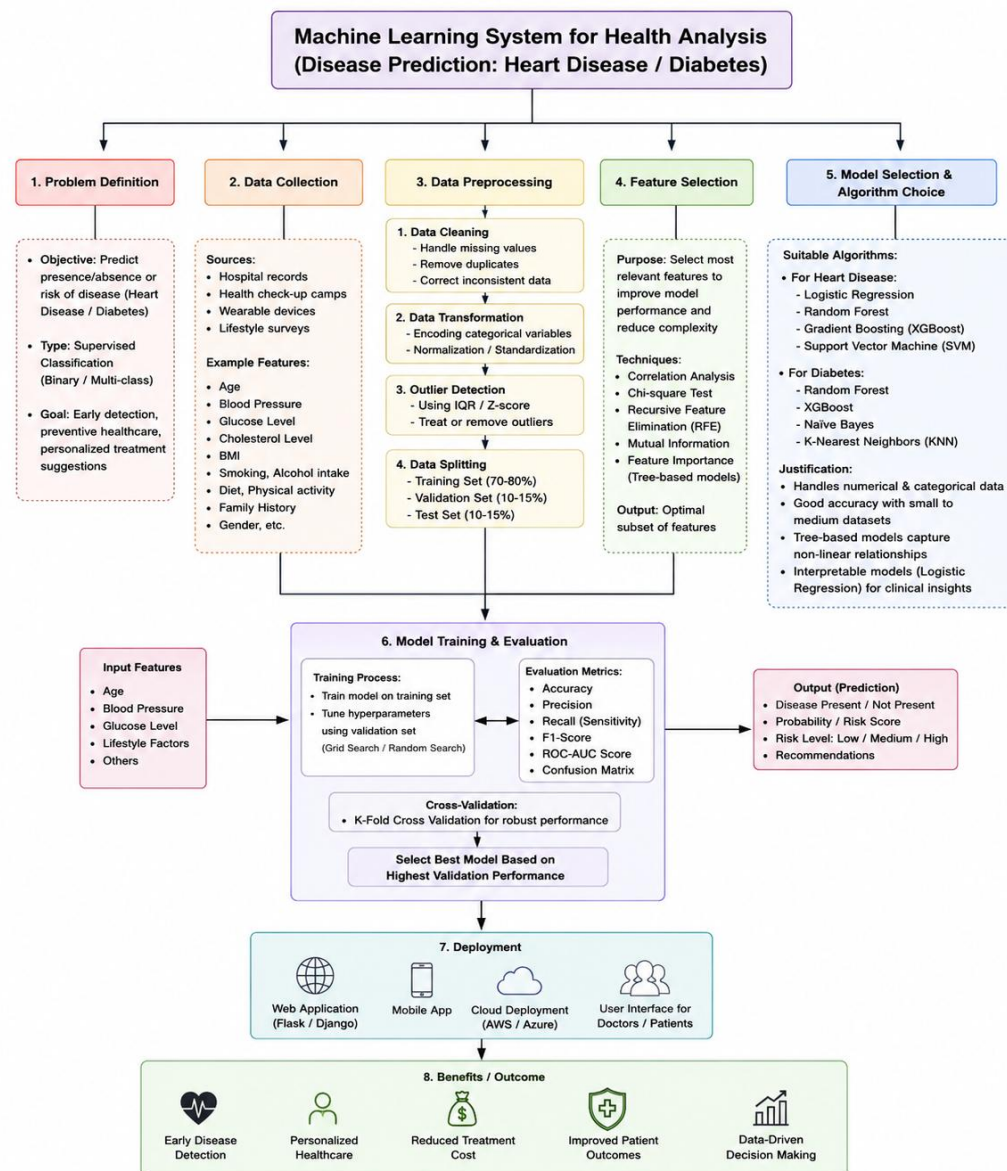
Course Code: ITA0609

Course Title: Machine Learning

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1. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions.



Machine Learning System for Health Analysis

Disease Prediction (Heart Disease / Diabetes)

1. Problem Definition

Purpose:

The first step is to define the objective of the system.

- The goal is to predict whether a patient has **heart disease or diabetes**.
- This is a **Supervised Machine Learning Classification** problem because the output is either **Disease Present** or **Disease Not Present**.
- Early prediction helps doctors provide timely treatment and improve patient care.

Input:

- Age
- Blood Pressure
- Glucose Level
- Lifestyle Factors

Output:

- Disease Present
- Disease Not Present

2. Data Collection

This stage collects patient information from different healthcare sources.

Possible data sources include:

- Hospital records
- Diagnostic laboratories
- Wearable health devices
- Electronic Health Records (EHR)
- Health surveys

Features collected

- Age
- Blood Pressure
- Blood Sugar (Glucose)
- Cholesterol
- BMI
- Smoking Habit
- Alcohol Consumption
- Physical Activity
- Family History

These features are used to train the machine learning model.

3. Data Preprocessing

Raw medical data usually contains missing or inconsistent values.

Before training the model, the data must be cleaned.

(a) Data Cleaning

- Remove duplicate records.
- Fill missing values.
- Correct incorrect entries.

Example:

Age = NULL

↓

Replace with average age.

(b) Data Transformation

Machine Learning algorithms require numerical values.

Categorical values are converted into numbers.

Example

Smoking

Yes → 1

No → 0

Blood Pressure

High → 2

Medium → 1

Low → 0

(c) Normalization

Different attributes have different ranges.

Example

Age = 65

Glucose = 180

These values are scaled to a common range (0–1).

Normalization improves prediction accuracy.

(d) Data Splitting

The dataset is divided into three parts.

Training Set → 70%

Validation Set → 15%

Testing Set → 15%

- Training data is used to learn patterns.
- Validation data is used to tune the model.
- Testing data evaluates final performance.

4. Feature Selection

Not every feature contributes equally.

Feature Selection identifies the most important features.

Example:

Important Features

✓ Age

✓ Blood Pressure

✓ Glucose Level

✓ BMI

✓ Family History

Less Important Features

✗ Patient ID

✗ Registration Number

These unnecessary attributes are removed.

Benefits

- Faster training
- Better accuracy
- Reduced complexity
- Less overfitting

5. Model Selection (Algorithms)

The next step is choosing the most suitable algorithm.

Logistic Regression

Used for binary classification.

Output

Disease

Yes / No

Why choose it?

- Easy to understand.
- Fast.
- Works well for medical diagnosis.

Random Forest

Random Forest creates many decision trees.

Final prediction is obtained using majority voting.

Advantages

- High Accuracy
- Handles large datasets
- Reduces overfitting
- Works well with medical data

Support Vector Machine (SVM)

SVM separates healthy and unhealthy patients using an optimal boundary.

Advantages

- Effective for small datasets.
- Good classification performance.
- High accuracy.

XGBoost

XGBoost combines many weak learners into one strong learner.

Advantages

- Very high accuracy.
- Handles complex medical relationships.
- Widely used in disease prediction research.

Why these algorithms?

These algorithms are selected because they

- Handle both numerical and categorical data.

- Produce high prediction accuracy.
- Reduce overfitting.
- Work efficiently with healthcare datasets.
- Can detect complex disease patterns.

6. Model Training

The selected algorithm is trained using the training dataset.

The model learns relationships such as

High Glucose

-

High Blood Pressure

-

High BMI

↓

Higher Diabetes Risk

Training continues until prediction error becomes minimum.

7. Model Evaluation

After training, the model is tested.

Performance Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix

These metrics determine whether the model performs well.

8. Prediction (Output)

The trained model predicts

Patient Data

↓

Machine Learning Model

↓

Disease Prediction

↓

Heart Disease = Yes/No

or

Diabetes = Yes/No

The system can also estimate

- Low Risk
- Medium Risk
- High Risk

and recommend that high-risk patients consult a doctor.

9. Deployment

Once the model is tested successfully, it can be deployed as:

- Hospital Management System
- Web Application
- Mobile Health App
- Cloud-based Healthcare Platform

Doctors can enter patient information and receive instant disease predictions.

10. Benefits

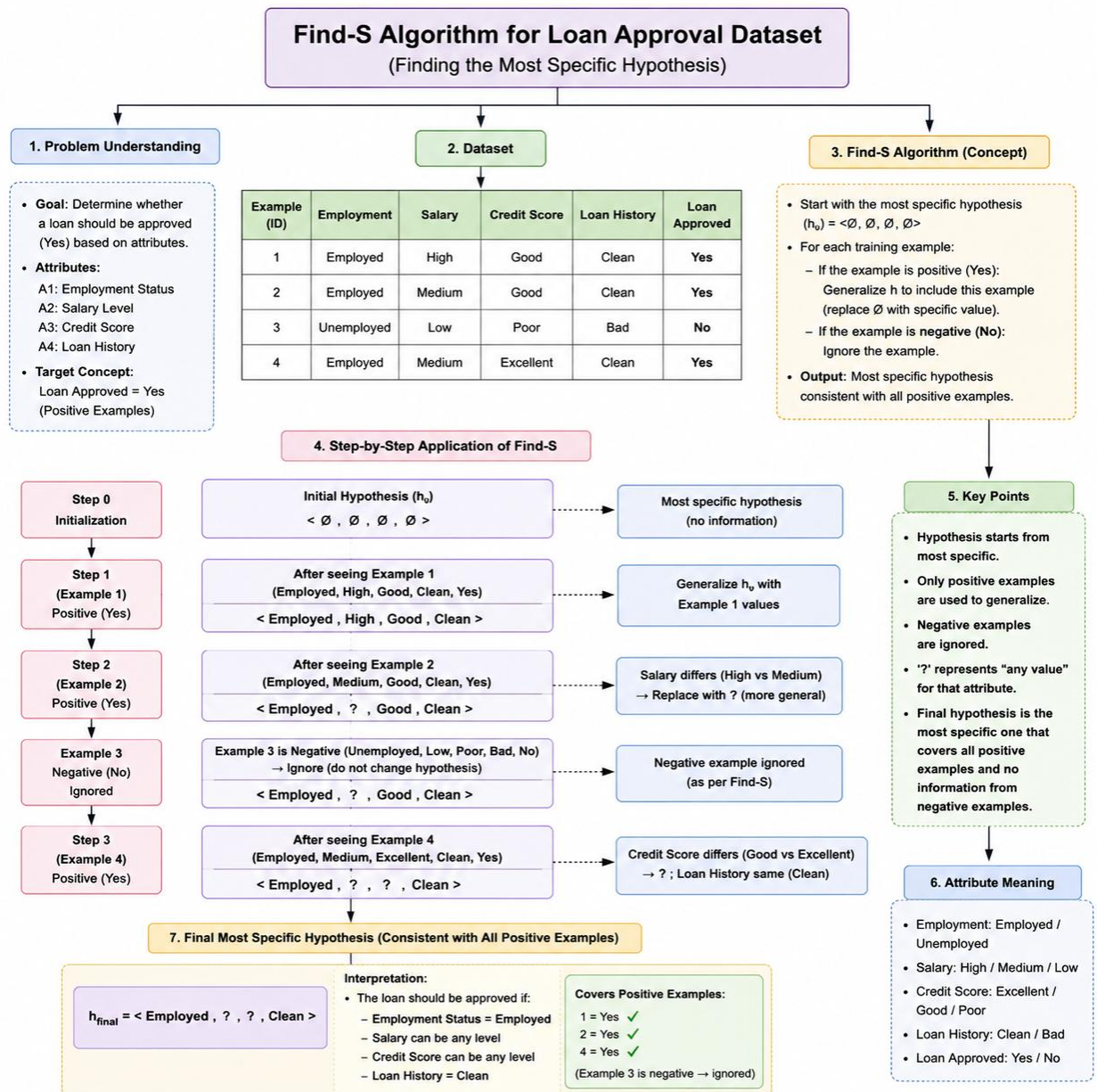
The proposed Machine Learning Health Analysis System offers several advantages:

- Early detection of heart disease and diabetes.
- Improved prediction accuracy using machine learning algorithms.
- Reduced diagnosis time for healthcare professionals.
- Personalized treatment recommendations.
- Better patient monitoring and preventive healthcare.
- Supports data-driven clinical decision-making.

Conclusion

The proposed Machine Learning Health Analysis System uses patient data such as age, blood pressure, glucose level, BMI, and lifestyle factors to predict diseases like heart disease and diabetes. After preprocessing the data and selecting the most relevant features, algorithms such as Logistic Regression, Random Forest, SVM, and XGBoost are trained to generate accurate predictions. This system helps healthcare professionals detect diseases early, improve diagnostic accuracy, and provide timely treatment, leading to better patient outcomes.

2. In the context of the **Find-S algorithm**, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples.



Find-S Algorithm for Loan Approval

1. Problem Understanding

The objective is to determine whether a **loan should be approved** based on the applicant's details. The target concept is **Loan Approved (Yes/No)** using attributes like Employment Status, Salary, Credit Score, and Loan History.

2. Dataset

The dataset contains four training examples with four input attributes:

- Employment Status
- Salary Level
- Credit Score
- Loan History

The output is **Loan Approved (Yes/No)**. Positive examples (Yes) are used to build the hypothesis.

3. Find-S Algorithm Concept

The Find-S algorithm starts with the **most specific hypothesis** and considers only **positive training examples**. Whenever a new positive example differs from the current hypothesis, the differing attribute is replaced with '?' (any value). Negative examples are ignored.

4. Step-by-Step Application

Step 0 – Initialization

The initial hypothesis is the **most specific hypothesis**:

$$h_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$$

This means that initially no attribute values are known.

Step 1 – Example 1 (Positive)

The first positive example is:

$\langle \text{Employed}, \text{High}, \text{Good}, \text{Clean} \rangle$

The hypothesis becomes:

$\langle \text{Employed}, \text{High}, \text{Good}, \text{Clean} \rangle$

because it is the first positive example.

Step 2 – Example 2 (Positive)

The second positive example is:

$\langle \text{Employed}, \text{Medium}, \text{Good}, \text{Clean} \rangle$

Only the **Salary** attribute differs (High $\rightarrow$ Medium), so it is generalized to ?.

Updated hypothesis:

$\langle \text{Employed}, ?, \text{Good}, \text{Clean} \rangle$

Step 3 – Example 3 (Negative)

The third example is negative:

$\langle \text{Unemployed}, \text{Low}, \text{Poor}, \text{Bad} \rangle$

Since Find-S considers **only positive examples**, this example is ignored and the hypothesis remains unchanged.

Step 4 – Example 4 (Positive)

The fourth positive example is:

⟨Employed, Medium, Excellent, Clean⟩

The **Credit Score** differs (Good → Excellent), so it is generalized to ?.

Final hypothesis:

⟨Employed, ?, ?, Clean⟩

5. Final Hypothesis

The most specific hypothesis consistent with all positive examples is:

h = ⟨Employed, ?, ?, Clean⟩

This means:

- Employment Status must be **Employed**.
- Salary can be **any value (?)**.
- Credit Score can be **any value (?)**.
- Loan History must be **Clean**.

6. Key Points

- Find-S starts with the **most specific hypothesis**.
- It **uses only positive examples** to generalize the hypothesis.
- **Negative examples are ignored**.
- The symbol '?' indicates that any value is acceptable for that attribute.

7. Conclusion

Using the Find-S algorithm, the final hypothesis is **⟨Employed, ?, ?, Clean⟩**. This hypothesis covers all positive loan approval examples while remaining as specific as possible, making it consistent with the given training data.